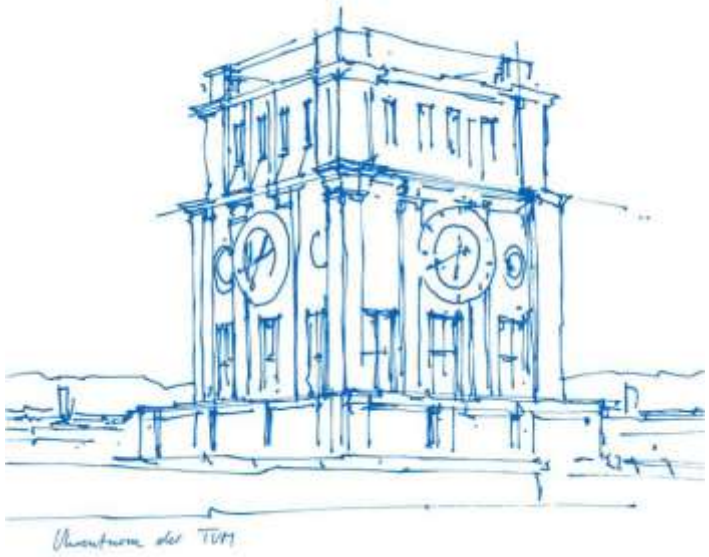
Hybrid Vision

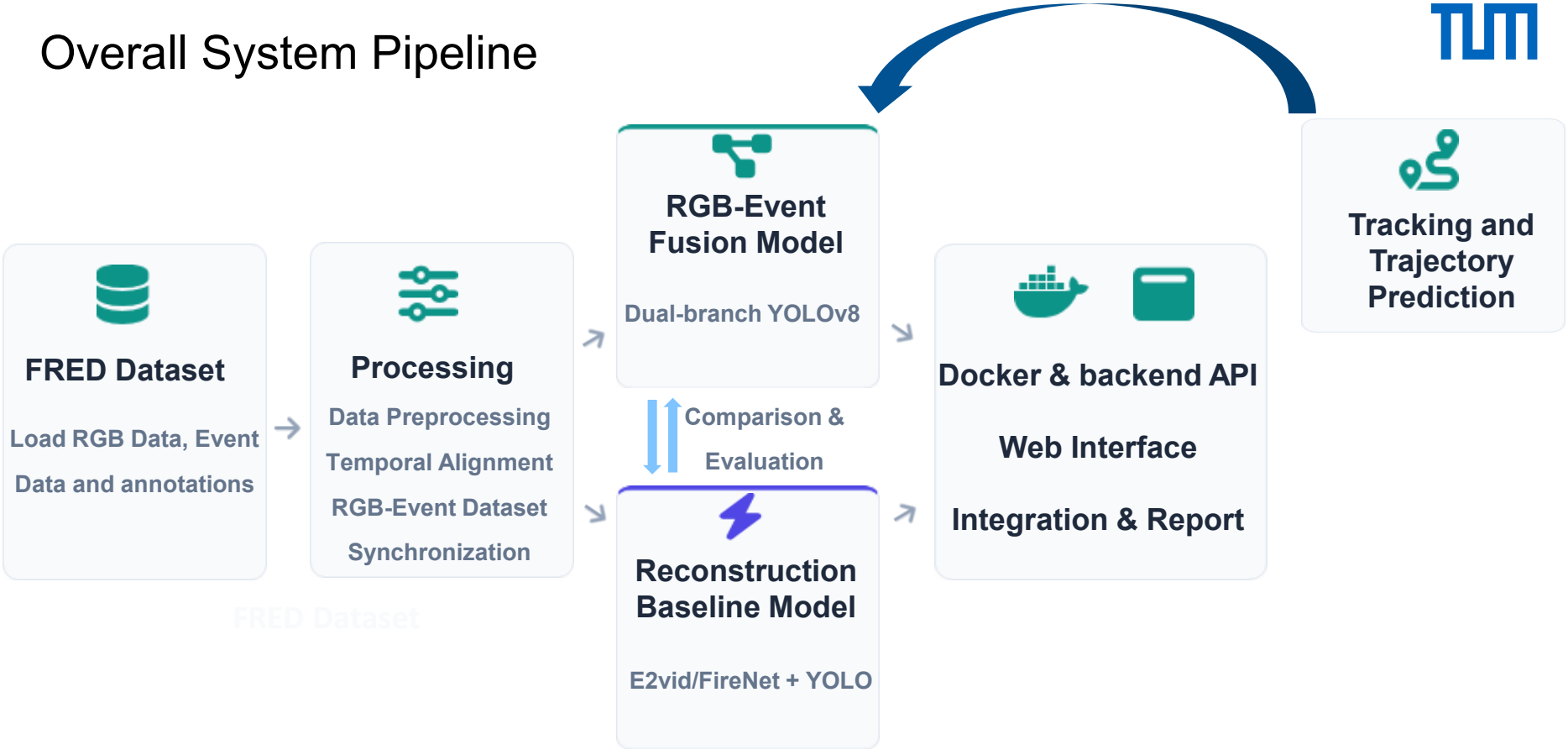
RGB + Event Drone Detection on FRED

Group 02
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Final Presentation
Technical University of Munich



Overall System Pipeline



Experimental Design

Follow the two-stage comparison:
first identify the best hybrid model, then compare it against reconstruction-based baselines.

1. Hybrid Model Comparison

Compared models:
- RGB-only
- Event-only
- Late Fusion: split into RGB branch, Event branch, and NMS fusion
- Early Fusion
- Two-Stream-Lite

Evaluation metrics:
- AP50
- Precision
- Recall

2. Best Model Selection

The best-performing hybrid model is selected based on overall detection performance (AP50, Precision and Recall).

Precision is reported separately.

3. Reconstruction Comparision

- E2VID + YOLO
- FireNet + YOLO

Research Question:

Can reconstructed event videos achieve comparable performance to hybrid vision?

Hybrid Models

All methods are trained for the same drone detection task. The difference is where RGB/Event information is fused.

RGB-only

Input:
RGB

Training:
Single detector

Fusion:
None

Output:
Detection

Event-only

Input:
Event

Training:
Single detector

Fusion:
None

Output:
Detection

Late Fusion

Input:
RGB + Event

Training:
Two detectors trained independently

Fusion:
Decision-level fusion (NMS)

Output:
Fused detections

Two-Stream-Lite

Input:
RGB + Event

Training:
Joint two-stream training

Fusion:
Feature-level fusion

Output:
Detection

Early Fusion

Input:
RGB + Event

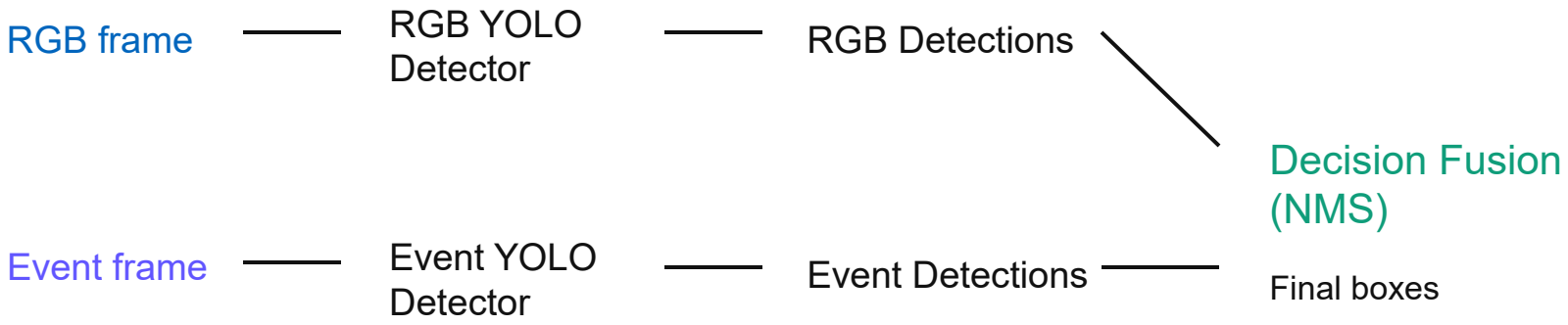
Training:
Single detector

Fusion:
Input-level fusion

Output:
Detection

Late Fusion (Decision-Level Fusion)

RGB and Event detectors are trained independently and fused using NMS.



Key point: RGB and Event do not interact during training; they are only merged after each branch has produced predictions.

Evaluation Pipeline

All hybrid models were evaluated using the same pipeline. The best-performing model was then selected for deployment.

Input package

Paired RGB/Event images

Ground-truth annotations

YOLO-format labels

Inference

For each frame:
1. Read RGB image
2. Read Event image
3. Concatenate to 6 channels
4. Run trained Two-Stream-Lite best.pt
5. Save prediction txt

Evaluation

Predictions are matched to labels with IoU >= 0.5.

TP/FP/FN are counted, then precision/recall and AP50 VOC11 are computed.

Hybrid Model Performance

Model	Precision	Recall	AP50	Output logic
RGB-Only	0.721	0.466	0.444	RGB YOLO
Event-Only	0.644	0.542	0.502	Event YOLO
Late Fusion NMS	0.662	**0.869**	0.739	2 branches + NMS
Early Fusion	0.816	0.787	0.777	6-ch input fusion
Two-Stream-Lite	**0.827**	0.787	**0.786**	2 stems + feature fusion

→ Two-Stream-Lite achieves the best overall performance

Key Findings

✓ Best AP50
Two-Stream-Lite

✓ Highest Precision
Two-Stream-Lite

✓ Selected for deployment
Two-Stream-Lite

Qualitative Comparison

Representative example from the validation subset.



- Green boxes are ground-truth labels
- Red boxes are model predictions
- IoU measures how well the prediction overlaps with the ground truth.

All models detect the drone, but Two-Stream-Lite achieves the tightest localization with the highest IoU.

Event-Only Reconstruction Baselines

Compare event-only reconstruction + YOLO with the selected hybrid RGB-event model.

Model	Precision	Recall	AP50
E2VID	0.000438	0.00528	0.00000421
FireNet	0.221	0.0797	0.0363

Key Observation

- Reconstruction pipelines provide a reference baseline.
- Direct RGB-event fusion achieved the strongest detection performance among the evaluated methods.
- Therefore, Two-Stream-Lite was selected for deployment in the web demo.

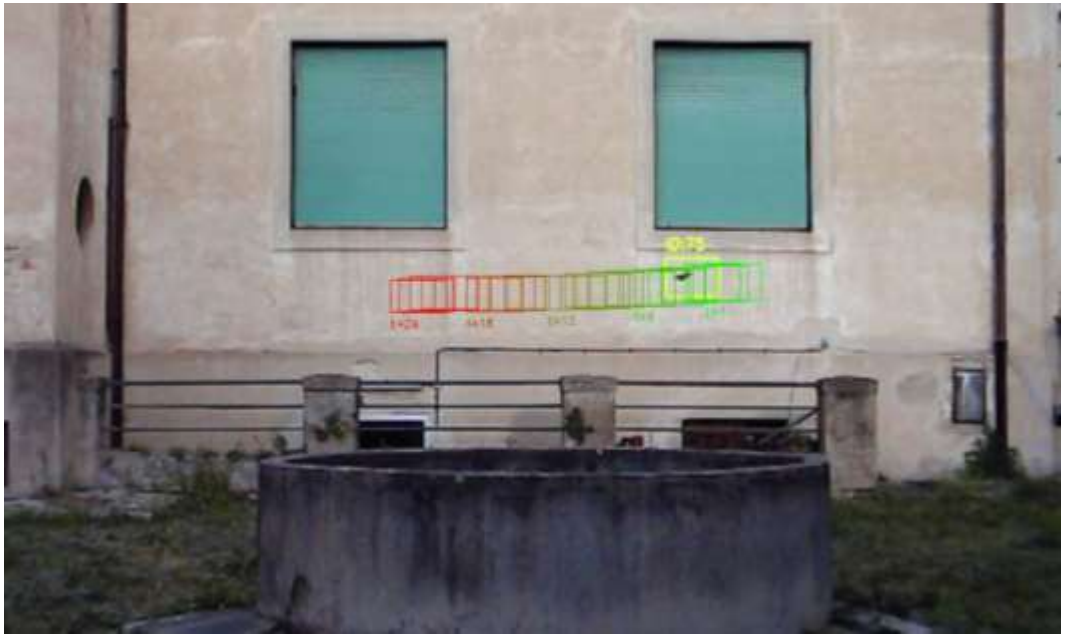
Tracking & Trajectory Prediction

We extend frame-level detection into temporal object tracking and short-term trajectory forecasting.

Tracking Performance

Precision	0.833
Recall	0.826
MOTA	0.656
ID Switches	15





Stable tracking (only 15 ID switches) with expected accuracy decay over the prediction horizon.

System Integration Pipeline

Integration of the selected detector into the end-to-end web application.

Selected Detector

Model
- Two-Stream-Lite

Input
- Paired RGB + Event frames

Output
- Bounding boxes
- Class labels
- Confidence scores

Export
- Standardized JSON
- Visualization images

Integration Layer

- Unified input format
- Standardized prediction API
- Consistent JSON output
- Model-independent interface

Deployment (Web Demo)

- Load detection results
- Synchronize RGB/Event playback
- Overlay predictions
- Interactive visualization

Take Away:
The selected Two-Stream-Lite detector was integrated through a standardized prediction interface, enabling seamless deployment in the Web demo.